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Raman spectroscopy of lithium hydride corrosion: Selection of appropriate excitation wavelength to minimize fluorescence

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ABSTRACT

Lithium hydride is pyrophoric and will efficiently react with moisture forming lithium hydroxide on the surface. Typically, this surface corrosion reaction is examined by diffuse reflectance infrared Fourier-transform (DRIFT) spectroscopy. In the current study, Raman spectroscopy has been used to evaluate the same reaction with a focus on the role of Li_2O in particular. A number of visible and near infrared excitation lasers were investigated in order to achieve the highest sensitivity; however, fluorescence dominates the spectrum. Photo bleaching experiments resulted in fluorescence reduction after illumination. Photoluminescence spectra indicate that the UV excitation could provide a luminescence-free region to collect high quality Raman spectra for LiH. A custom Raman probe was used to study the hydrolysis reaction of LiH with a 325 nm laser. Further, the role of lithium oxide in the hydrolysis kinetics has been studied.

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1. Introduction

Lithium hydride is a light, hydrogenous material which is used as a reducing agent and is being considered as a potential hydrogen storage material [1,2]. The lightest of molecular solids, LiH is the highest density form of hydrogen storage. The thermal dehydrogenation temperature of lithium hydride is too high to meet US DOE automotive targets [3] by thermal hydrogen release alone, however, some have considered a hydrolysis approach by which LiH is reacted with water in a controlled fashion to release hydrogen fuel [1,2]. LiH does play a vital role in amide [4–6], alanate [7], and borohydride [8] based storage systems, where it appears as a dehydrogenation product. In the amide system, it has been shown that LiH plays a role in scavenging NH_3 formation which lowers hydrogen capacity and reversibility and has been used as an additive to improve hydrogen storage performance [5]. If, however, LiH is exposed to water vapor, scavenging ability decreases as an exothermic reaction occurs resulting in caustic LiOH formation and hydrogen gas [6].

The high chemical reactivity of LiH in the presence of even minor concentrations of water makes handling and storage difficult from both a quality and safety standpoint. Maintaining pristine LiH is impossible despite handling the material under dry, inert atmospheres [9]. Thus, long-term storage of LiH hydrogen generation becomes a concern [1,2].

A variety of spectroscopic techniques [1,2,4,9–21] have been used to study surface corrosion reactions as well as their kinetics. Diffuse Reflectance Infrared Fourier Transform (DRIFT) spectroscopy is the most widely used technique to study the hydrolysis reaction on a lithium hydride surface [10,22–27]. The LiOH symmetric stretching mode grows with the addition of water vapor in a clear region of the spectrum. After hydrolysis, hydroxide can be removed by heating under vacuum to form Li_2O [28]. The role of Li_2O in the vigorous kinetics is not, however, well understood since the lone infrared active Li_2O phonon mode is obstructed by other features in the spectrum. As a complementary technique to infrared, Raman spectroscopy seems to be an ideal method to study the role of Li_2O—which is Raman active—in the hydrolysis reaction.

Charton et al. reported the first Raman spectroscopic study of surface hydrolysis by moisture vapor on LiH powder [29,30]. The Raman spectrum of LiH, LiOH, $LiOH{\cdot}H_2O$, and Li_2O pure powders have been reported previously [31–34]; however, Charton et al. [29,30] point out that consistent observation of hydrolysis products on the surface of lithium hydride powder is complicated by the presence of photoluminescence in the spectrum.

In the present study, a series of excitation wavelengths in the UV, visible, and near-IR are investigated to determine if Raman spectroscopy is a viable method for the measurement of surface hydrolysis on lithium hydride. The purpose herein is to determine the best practical system for investigation of moisture reactions with LiH rather than identification of the spectrometer with the best absolute sensitivity. The interplay between fluorescence of LiH and Raman scattering intensity, which increases with shorter excitation wavelengths is the critical parameter.

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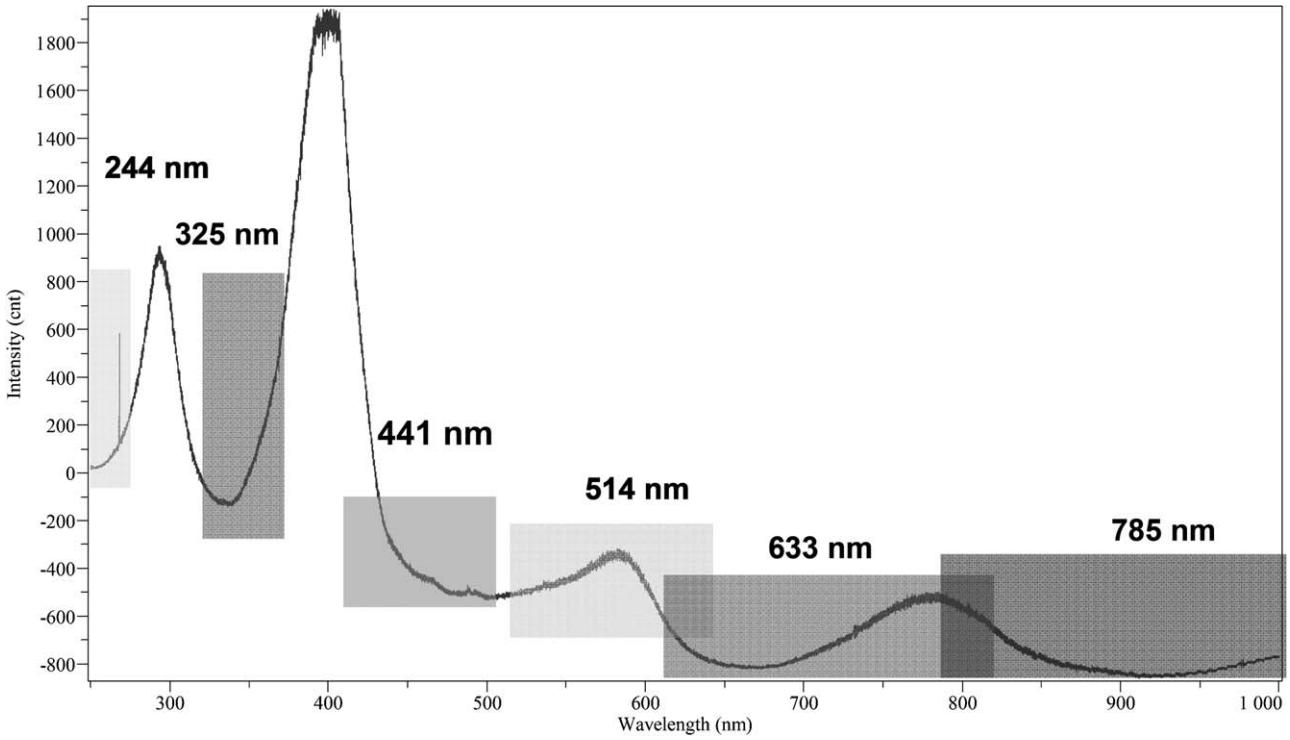


Fig. 1. Photoluminescence spectrumof corroded a compacted lithium hydride sample. The shaded areas indicate the Raman scattering region (0–4000 cm^{-1}) for a given excitation wavelength.

2. Experimental

Multiple Raman spectrometers are used which allows for measurement of the same set of lithium hydride samples at a multitude of excitation wavelengths. These spectrometers include a NXR FT-Raman module (1064 nm) attached to a Thermo-Nicolet model 6700 FTIR. A second spectrometer system consists of a Horiba model T64000 and utilizes several visible wavelengths. The third system is a Renishaw spectrometer (model RX-500) fitted with fiber optics delivery and collection of the incident laser light and subsequent Raman scattering and utilizes a 325 nm UV laser. In total, 244, 325, 442, 488, 514, 532, 660, 676, 785, and 1064 nm lasers were tested. Microscope configurations were typically used with a 20× focusing objective. Sampling areas of 50 μm to 1 mm were investigated with little change in the observed LiOH signal-to-fluorescence ratio.

A series of lithium hydride samples was prepared by pressing LiH powder into a solid disk (13 mm diameter × 2 mm thick). Each sample was exposed to H_2O vapor entrained in an argon gas stream for differing periods of time. The thickness of the LiOH corrosion layer resulting from reaction of LiH with H_2O in the moisture stream was confirmed by Diffuse Reflectance Infrared Fourier Transform (DRIFT) spectroscopy. The LiOH thickness on the lithium hydride surface is correlated to the infrared absorbance based on methodology developed by Powell and Barber [23]. Samples ranged in corrosion from less than one nanometer LiOH thickness on the LiH surface to approximately 1 micron.

A highly corroded lithium hydride sample (LiOH thickness ≫1 μm) was placed in an evacuable cell [22] with a quartz window on it in order to measure Li_2O formation in situ. When heated under vacuum to 350 °C, LiOH will decompose to Li_2O and hydrogen gas. The UV laser was focused on the sample surface through the quartz window and spectra were collected every minute during heating.

3. Results and discussion

A variety of excitation laser choices exist in the UV, visible and near-IR region of the light spectrum. As has been indicated previously, fluorescence was observed in the Raman spectrum of corroded lithium hydride [29,30] which limited the usefulness of the Raman technique. In order to determine the best excitation wavelength, a photoluminescence spectrum (Fig. 1) was collected on a lithium hydride disk with approximately 200 μm of LiOH on the surface. No Li_2O was present. In the spectrum, large photoluminescence bands are observed. The shaded areas within the spectrum represent the Raman shift region (0–4000 cm^{-1}) for a series of excitation wavelengths. It can be seen that some wavelengths are not ideal for investigation of lithium hydride due to fluorescence bands. The UV wavelengths and 785 nm appear to be reasonable laser probes. 1064 nm will also be investigated as it is commonly used to collect Raman spectrum of samples which fluoresce strongly at visible wavelengths.

Raman spectra of five corroded lithium hydride samples are shown in Fig. 2 (λ = 785 nm). While the LiOH concentration varied significantly, the spectrum features did not. The lithium hydride vibration at 492 cm^{-1} was consistently observed; however, no other features representative of LiOH were observed. Fluorescence dominates all of the spectra. Attempts to apply signal processing routines to enhance Raman scattering signals were unsuccessful.

Near IR lasers are often used to overcome sample fluorescence; however, a tremendous reduction in Raman intensity occurs. Approximately 1% of all excitation photons interact with the sample causing inelastic (Raman) scattering. The Raman scattering intensity is inversely proportional to the fourth power of the laser wavelength such that longer wavelengths result in significantly weaker Raman spectra. Lithium hydride samples were interrogated with a FT-Raman microscope instrument operating at 1064 nm as shown by the dashed spectrum in Fig. 3. A broad

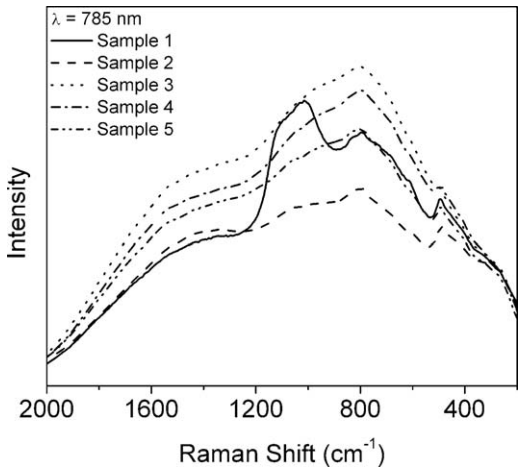


Fig. 2. Raman spectra of five LiH samples with varying LiOH surface concentrations collected with a 785 nm laser. Fluorescence dominates the spectra.

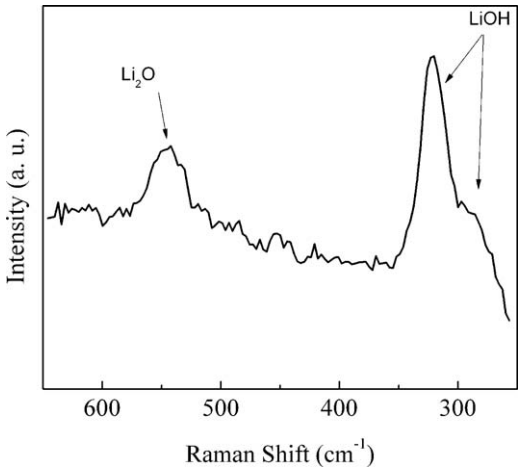


Fig. 4. UV Raman spectrum of a compacted LiH sample with a Li_2O surface layer after moisture exposure. The Li_2O vibration is clearly observed at 515 cm^{-1}.

fluorescence feature from 1000 to 3000 cm^{-1} dominates the spectrum. Small lithium hydroxide vibrations are observed at 323, 610, and 3670 cm^{-1} as expected. The Raman intensity for these LiOH bands was weak despite the sample being highly corroded. Photobleaching was also attempted in an effort to remove fluorescence. This is a method in which the laser photons irradiated the sample constantly for a long period of time. The Raman scattering features of the spectrum will not change due to this extended irradiation; however, fluorescence will decrease. The net result is concentration of the Raman spectral features by reducing the overall fluorescence background. In the current study, a sample was irradiated for 45 min resulting in the background being reduced by 60–70% without further resolution of any discernable Raman features. Burning the sample during photobleaching due to the long term laser exposure is also a concern. The intense localized laser energy can heat the sample changing the surface chemistry.

It is important to note that the fluorescence observed varied greatly among samples analyzed. The origin of the observed fluorescence is not understood, but Anderson and Lüty suggest that it lies in defects site within the lithium hydride lattice [31]. These defects are due, according to Anderson and Lüty, to non-stoichiometric hydrogen regions within LiH crystallites. When insufficient hydrogen is present, high luminescence is observed in the Raman spectrum. When the authors apply an overpressure of hydrogen to the sample, the observed luminescence feature disappears. Variation in the current LiH compressed samples would then be due to crystallites within the compressed solid having differing amounts of slight non-stoichiometry with regard to the hydrogen concentration.

With visible and near-IR laser excitation suffering from significant fluorescence interference, UV excitation was attempted. Both 244 nm and 325 nm lasers were successfully used to collect fluorescence free corrosion spectra from the lithium hydride surface. The solid spectrum in Fig. 3 shows the 325 nm UV Raman spectrum of the same sample. A much stronger scattering signal is observed with minimal fluorescence. The photoluminescence band observed in Fig. 1 begins just outside the scattering region of interest. UV excitation is the best laser wavelength for Raman investigations of lithium hydride corrosion.

With fluorescence contributions to the Raman spectrum mitigated, a lithium hydride solid was conditioned such that the surface had a Li_2O surface layer prior to exposure to moisture. This is accomplished by heating the sample under vacuum to 350 °C after an initial water exposure to drive LiOH to Li_2O. The LiH sample with a Li_2O surface layer is then cooled and further exposed to more water vapor in order to hydrolyze the surface again. A Raman band, characteristic of Li_2O, is observed in the phonon region along with LiOH features at 323 and 280 cm^{-1} (Fig. 4). In prior reports, fluorescence interference or low scattering intensity have limited the ability to observe Li_2O and elucidate the role it plays in moisture

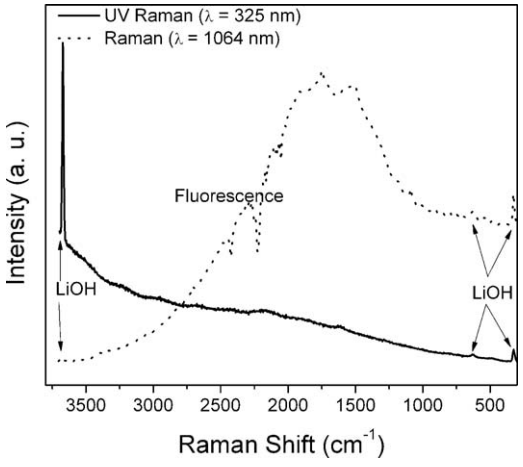


Fig. 3. Raman spectra of a highly corroded lithium hydride sample collected with a 1064 nm laser (dotted line) and at 325 nm (solid line). Fluorescence is minimized by employing a UV excitation laser.

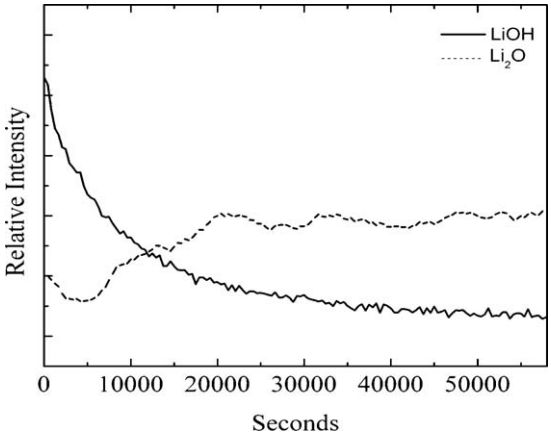


Fig. 5. Raman scattering intensity plot for LiOH and Li_2O as a function of vacuum heating time. LiOH decomposes to Li_2O under these conditions.

corrosion. UV Raman spectroscopy now provides a direct method to study the LiH/Li_2O reaction with water. Fig. 5 presents the Raman intensity of LiOH and Li_2O as a function of time during heating under vacuum. As the corroded LiH solid is heated, LiOH decomposes into Li_2O and hydrogen gas such that the Li_2O signal increases in time. Further experiments are being conducted to compare the kinetics of the reaction with previous reports [18].

4. Conclusions

Lithium hydride is used in a variety of industries; however, its chemical reactivity with water makes safe handling and storage a challenge. The presence of LiOH, as a result of hydrolysis, typically reduces the usefulness of LiH in these industrial applications. Understanding the kinetics of this reaction, and factors that control it is therefore important. Infrared spectroscopy has been extensively used to study the chemistry; however, direct investigations of the role of Li_2O are not possible. Raman spectroscopy is an excellent complementary method since Li_2O vibrations are Raman active. Employing UV laser wavelengths to excite the sample overcomes fluorescence interferences while allowing all of the spectral features from LiH, LiOH, and Li_2O to be observed.

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